

COURSE SYLLABUS

Advanced Automated Administration With PowerShell

COURSE NUMBER

55318B

DURATION

3 days · 9 modules

AUDIENCE LEVEL

Advanced

FORMAT

Instructor-Led / Self-Paced

Course syllabus

COURSE OVERVIEW

This three-day course teaches experienced IT professionals how to build reliable, reusable, and production-ready automation with modern PowerShell. The course updates the previous Windows PowerShell-focused curriculum for current corporate environments that use PowerShell 7.x, Windows Server 2025, SQL Server 2025, REST APIs, Git-based workflows, and secure automation. Students create advanced functions, orchestrate complex workflows, manage errors securely, and integrate AI-assisted analysis tools.

AT COURSE COMPLETION

After completing this course, you will be able to:

Design and deploy production-grade advanced functions, script modules, and controller scripts using modern PowerShell 7.6.

Integrate external interfaces, call REST APIs, parse JSON payloads, and retrieve datasets from SQL Server 2025 databases safely.

Implement enterprise exception trapping, system event logging, task scheduler execution, and multi-threaded background jobs.

Establish robust security controls, protect administrative secrets, build JEA endpoints, and structure projects using Git version control.

Utilize modern testing, static debugging tools, and AI validation methods to inspect automated code structures safely.

PREREQUISITES

- Experience with administrative scripting in Windows PowerShell 5.1 or PowerShell 7.x
 - Familiarity with basic programming concepts such as variables, loops, and conditions
 - Basic understanding of SQL Server and database query concepts
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COURSE OUTLINE

MODULE 1 Creating advanced functions

120 min

Convert repeatable script logic and simple commands into robust, professional-grade advanced functions.

LESSONS

- Lesson 1:** Defining Parameters, Validation, and Pipeline Inputs
- Lesson 2:** Integrating Safety Features and AI Review Workflows

LAB EXERCISES

- Creating an Advanced PowerShell Function for SQL Server Auditing

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Create reusable advanced functions that mirror compiled cmdlets
- Define parameters with advanced validation attributes and pipeline binding
- Generate consistent object-oriented output structures
- Add comment-based help and support verbose logging and debugging modes
- Support WhatIf and Confirm risk mitigation and validate code using AI assistance

MODULE 2 Using .NET and REST APIs in PowerShell

120 min

Extend standard PowerShell administrative automation by interacting directly with the .NET framework, remote REST web services, and SQL Server 2025 databases.

LESSONS

- Lesson 1:** Using .NET Classes and Methods in PowerShell
- Lesson 2:** Interacting with REST APIs and Parsing JSON Data
- Lesson 3:** Querying SQL Server 2025 Databases

LAB EXERCISES

- Calling Web Services and Fetching AdventureWorks2025 Records

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Incorporate .NET classes and static methods into PowerShell scripts
- Execute REST API web calls using Invoke-RestMethod and Invoke-WebRequest
- Process HTTP request headers, security tokens, and JSON data structures
- Extract database records from SQL Server 2025 utilizing the SqlServer module
- Validate AI-suggested query logic and isolate external integration failures

MODULE 3 Writing controller scripts

120 min

Design and build high-level orchestration controller scripts that integrate standalone custom functions with databases, reports, and API sources.

LESSONS

- Lesson 1:** Architecture of PowerShell Controller Scripts
- Lesson 2:** Integrating Diverse Inputs and Sequencing Functions
- Lesson 3:** Generating Structured Reports and Markdown Documentation

LAB EXERCISES

- Orchestrating Workflows and Automated Report Generation

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Differentiate between reusable script modules and high-level controller processes
- Implement master scripts that query data from files, APIs, and databases
- Coordinate dependent utility functions to handle complex workflows
- Generate automated operational reports in Markdown, CSV, and structured formats
- Review architectural script layouts with AI tools to identify structural issues

MODULE 4 Handling script errors

120 min

Build high-resiliency automation scripts that fail gracefully, log useful diagnostics, and apply automated recovery routines.

LESSONS

- Lesson 1:** Error Types, Terminology, and Action Behaviors
- Lesson 2:** Defensive Scripting with Try-Catch-Finally Blocks
- Lesson 3:** Implementing Robust Logging, Retries, and Timeouts

LAB EXERCISES

- Building Exception-Tolerant Scripts with Active Logging

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

- Distinguish terminating errors from non-terminating errors
- Control error-handling behavior using ErrorAction variables
- Employ Try, Catch, Finally, and Throw blocks to manage exceptions
- Implement recovery paths, dynamic retries, and network timeout loops
- Leverage AI systems to evaluate and identify edge-case exceptions

MODULE 5 Working with structured data and automation reports

120 min

Read, validate, transform, and export CSV, JSON, XML, and SQL database outputs to generate complex enterprise reports.

LESSONS

Lesson 1: Ingesting, Parsing, and Validating CSV, JSON, and XML

Lesson 2: Designing Dynamic Structured Reports and Output Validations

LAB EXERCISES

Parsing Data Feeds and Compiling AdventureWorks Reports

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Import, validate, and parse structured CSV, JSON, and XML data files

Extract, transform, and map SQL Server 2025 result datasets

Clean, normalize, and format structured inputs dynamically

Compile automated Markdown and custom HTML status summaries

Use AI validation models to check schema logic and identify data transformation assumptions

MODULE 6 Enhancing server management with configuration and constrained administration

120 min

Implement configuration management patterns and secure, delegated system administration using modern frameworks and least-privilege concepts.

LESSONS

Lesson 1: Modern Desired State Configuration (DSC) as Code

Lesson 2: Implementing Just Enough Administration (JEA) Endpoints

Lesson 3: Analyzing Security Controls and Least-Privilege Frameworks

LAB EXERCISES

Mocking Constrained Configurations and Least-Privilege Audits

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Contrast configuration-as-code patterns with manual administrative patterns

Differentiate between Desired State Configuration (DSC) and Just Enough Administration (JEA)

Audit permission scopes and target credentials across APIs, files, and SQL engines

Evaluate use cases for DSC, GPO, Azure Policy, Intune, Azure Arc, and custom security scripts

Leverage AI systems to evaluate configuration-management options and access controls

MODULE 7 Analyzing and debugging scripts

120 min

Debug run-time scripting bugs, perform static code analysis, and run automated testing frameworks with Visual Studio Code.

LESSONS

Lesson 1: Static Code Analysis using PSScriptAnalyzer

Lesson 2: Introduction to Unit Testing with Pester Framework

LAB EXERCISES

Debugging Complex Flows and Authoring Basic Pester Tests

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Debug complex administrative scripts within Visual Studio Code

Leverage breakpoints, variable watches, and code stepping controls

Analyze automation files for styling, performance, and security issues using PSScriptAnalyzer

Author simple Pester 5.x unit and infrastructure validation tests

Audit script logic for hallucinated modules and errors using AI analysis tools

MODULE 8 Running and orchestrating PowerShell automation

120 min

Schedule, execute, and scale PowerShell automation tasks securely across local servers, pipelines, and enterprise runners.

LESSONS

Lesson 1: Managing Background Jobs, Threading, and Remote Workloads

Lesson 2: Enterprise Orchestration via Runbooks and GitHub Pipelines

Lesson 3: Managing Data Dependencies and Unattended Execution Environments

LAB EXERCISES

Scheduling Unattended Jobs and Caching Remote SQL Data

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Orchestrate script jobs using local background instances and secure PSRemoting paths

Describe cloud automation capabilities including Azure Automation runbooks

Explain CI/CD infrastructure automation strategies including GitHub Actions and Azure Pipelines

Incorporate data-source caching into offline, unattended script runs

Evaluate orchestration platforms using AI-assisted architectural analysis

MODULE 9 Building production-ready PowerShell automation

120 min

Structure, package, secure, and document advanced automation scripts to ensure they are production-ready and maintainable.

LESSONS

Lesson 1: Source Control Architecture and Project Structuring in Git

Lesson 2: Securing Secrets, Credentials, Execution Policies, and Code Signing

Lesson 3: Operational Documentation, Release Checklists, and AI Security

LAB EXERCISES

Packaging Modules, Managing Git Workflows, and Hardening Secrets

AFTER COMPLETING THIS MODULE, YOU WILL BE ABLE TO

Structure dynamic scripting folders as formal Git project workspaces

Enforce access controls, execution security, credential handling, and code signing

Document integrations, database connections, and external resource paths

Create standard handover materials, runbooks, and implementation steps

Establish AI policies and complete an architectural review of your final automation code